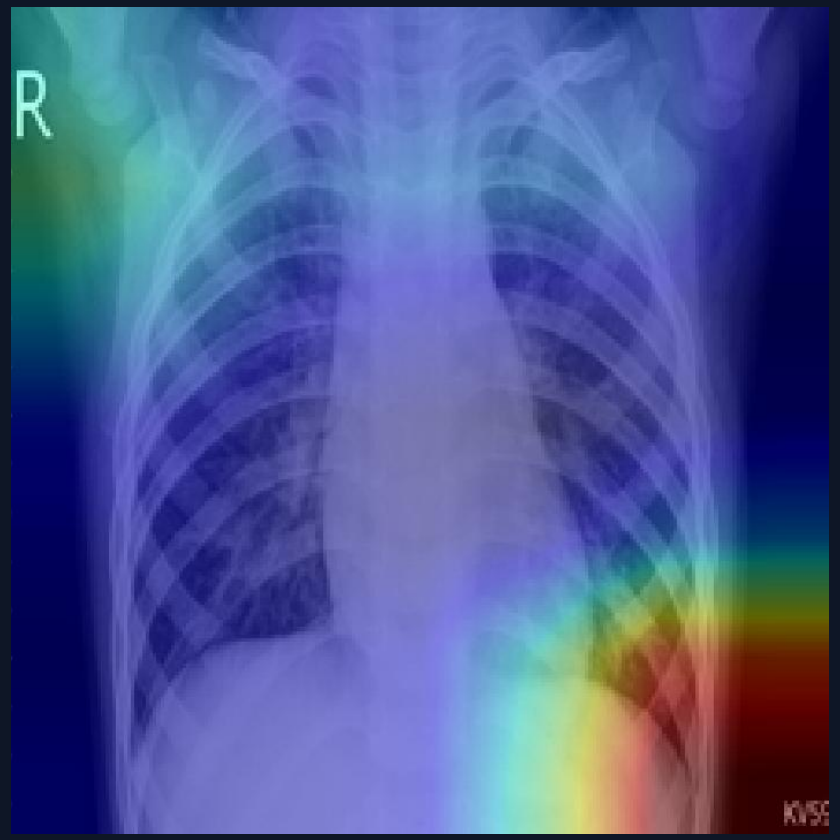
Is the model looking at the lungs?

Classical ML vs deep learning for paediatric pneumonia detection: evaluation, explainability and ethics



Healthy child, scored as near-certain pneumonia
Grad-CAM heat sits on the ‘KV59’ label, not the lungs

Machine Learning

Pneumonia is a leading killer of young children

740,180

children under 5 died of pneumonia in 2019, 14% of all under-5 deaths

Source: WHO, Pneumonia in children fact sheet (2022)

Research question

Which is better at supporting radiologists in triaging children's chest X-rays, a classical SVM or a deep CNN? And can we trust why it decides?

01 **Data**

02 **Models**

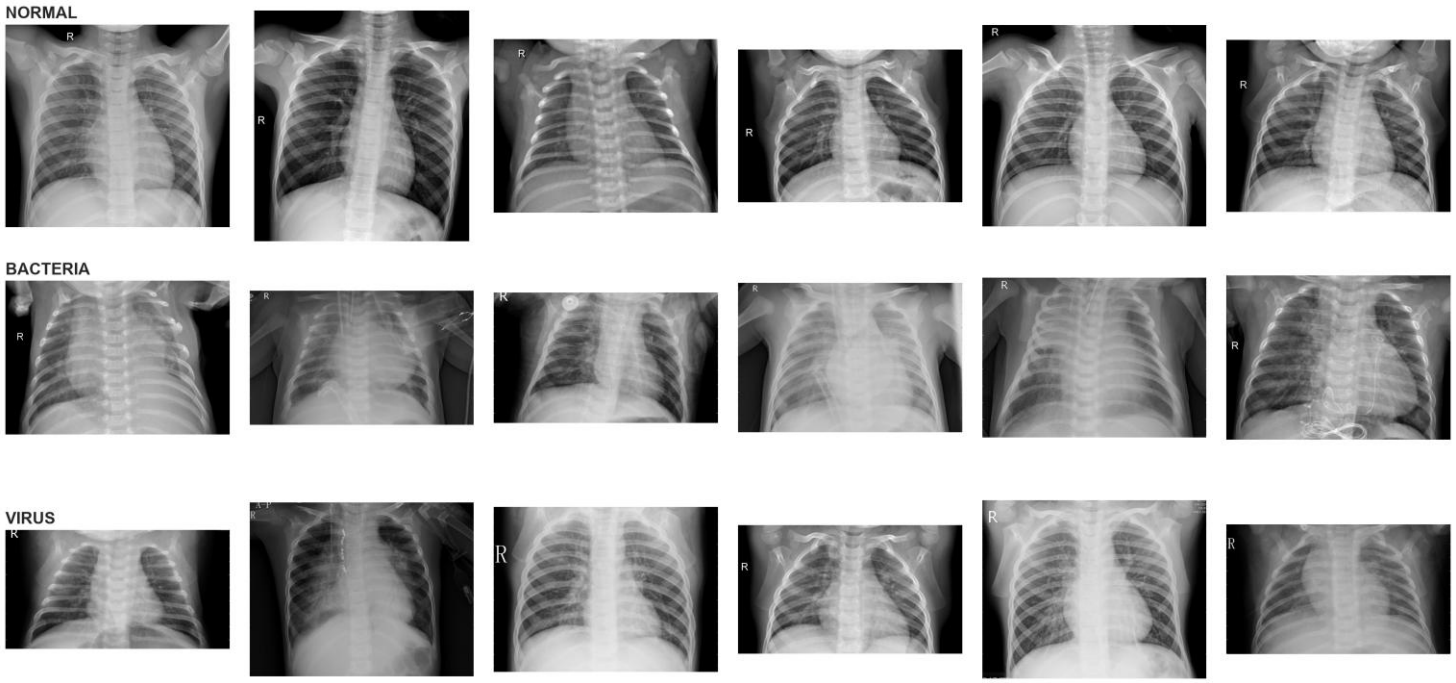
03 **Validation**

04 **Explainability**

05 **Bias and ethics**

06 **Deployment**

A real clinical dataset from one hospital



5,856

chest X-rays (normal, bacterial, viral)

≈ 4,100

children (patient IDs from filenames), aged 1–5

1 hospital

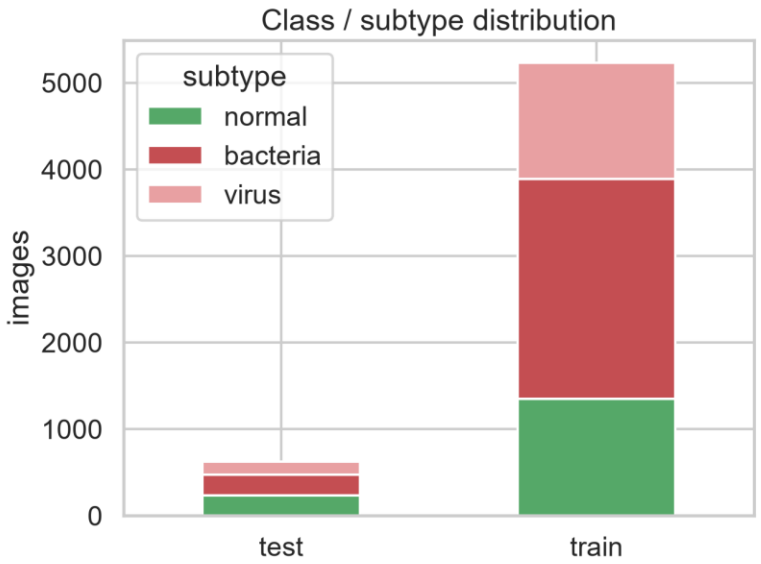
Guangzhou Women & Children's Medical Center

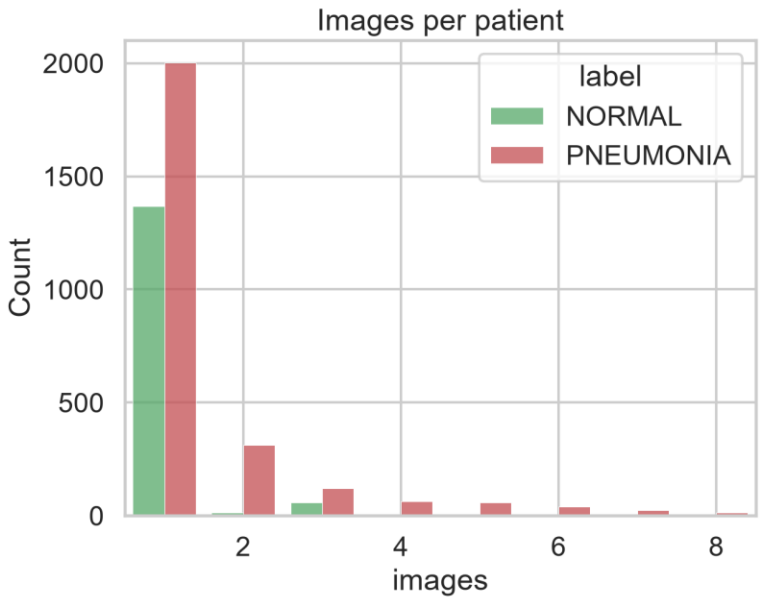
CC BY 4.0

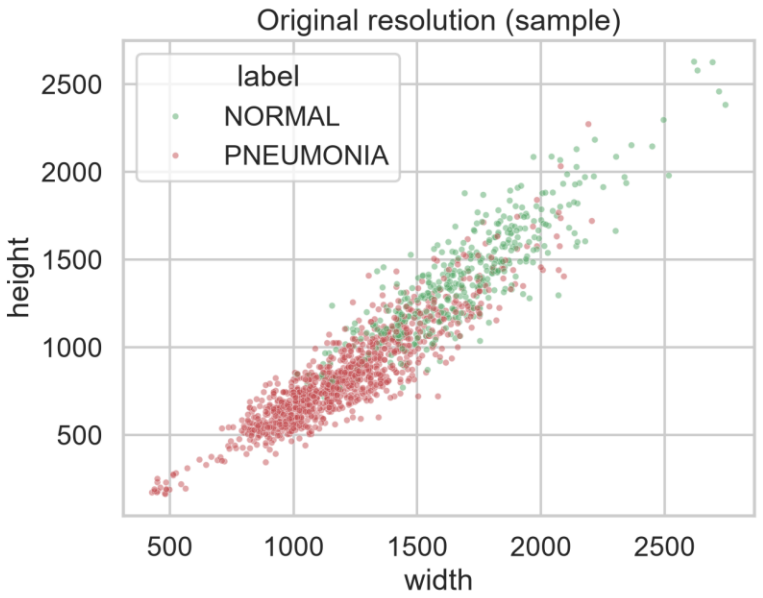
Kermany et al., Cell (2018), Mendeley Data

Why this dataset: real clinical stakes · patient IDs allow leakage-free validation · bacterial/viral labels allow subgroup analysis

What the data told us before modelling







2.88 : 1 imbalance

→ Class-weighted loss; judge by precision, recall, F1

Several images per child

→ Group folds by patient to stop leakage

Normal images ≈ 2.4× the pixels

→ Greyscale, resize to 224×224; watch for confounding

Burnt-in text and markers

→ Check what the models attend to with XAI

Two models, same data, same folds

Classical ML

scikit-learn

Resize to 128×128 → HOG edges (1,764) + intensity histogram (32) → Standardise → PCA, 95% variance (455 dims) → **RBF SVM · C = 10, γ = scale · class-balanced**

Hand-crafted edge and texture features: strong on small data, fast on CPU, but blind to anything HOG cannot describe

Deep learning

PyTorch

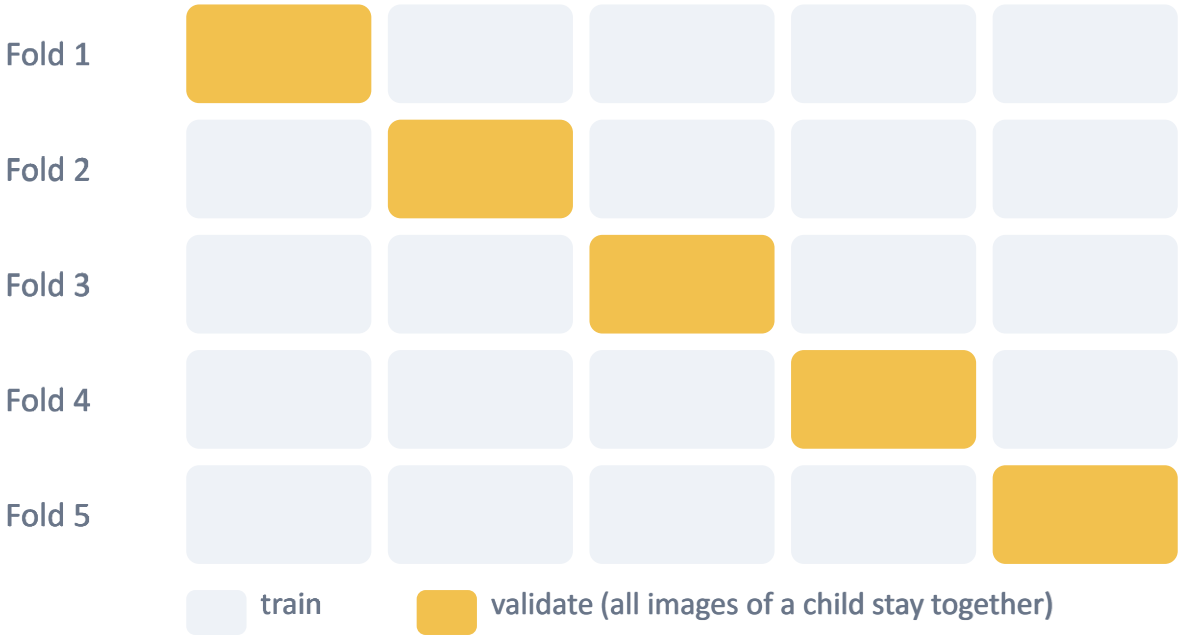
Resize to 224×224 → GPU augmentation: ±10° rotation, shift, zoom, exposure · no flips → ResNet-18, ImageNet weights, all layers fine-tuned → **AdamW 1e-4, cosine · weighted loss · early stopping (7 epochs)**

Learns its own features; transfer learning offsets the small dataset; 11.2M parameters, so it needs validation to catch overfitting

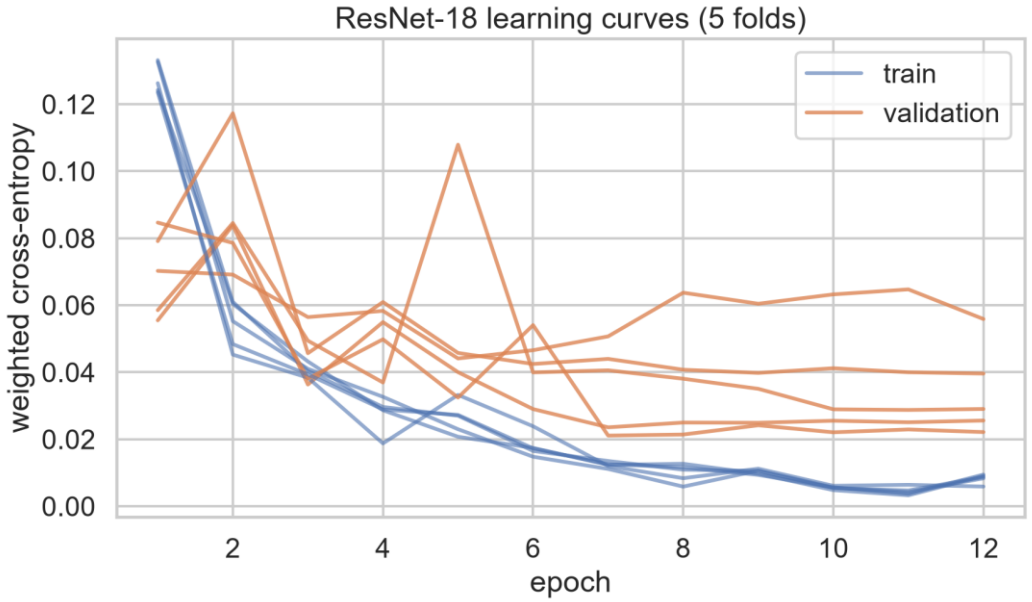
Shared: identical patient-grouped folds · identical held-out test set · identical evaluation code

Validating on new patients, not new images

Training set: 5,232 images · ≈ 3,700 children

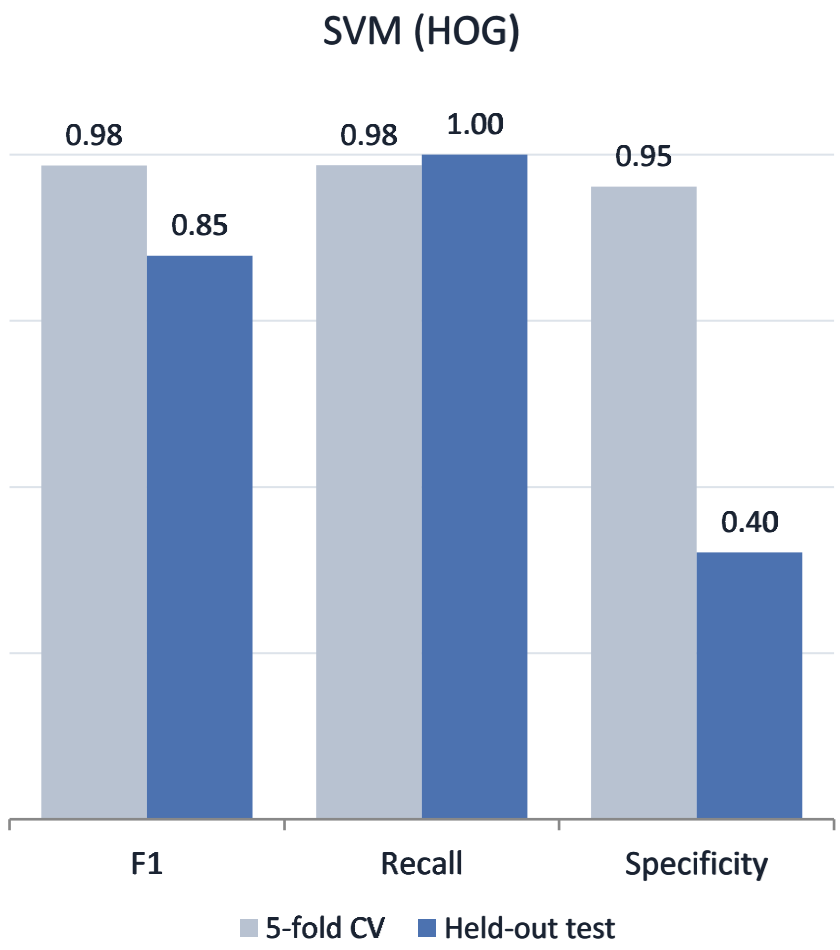


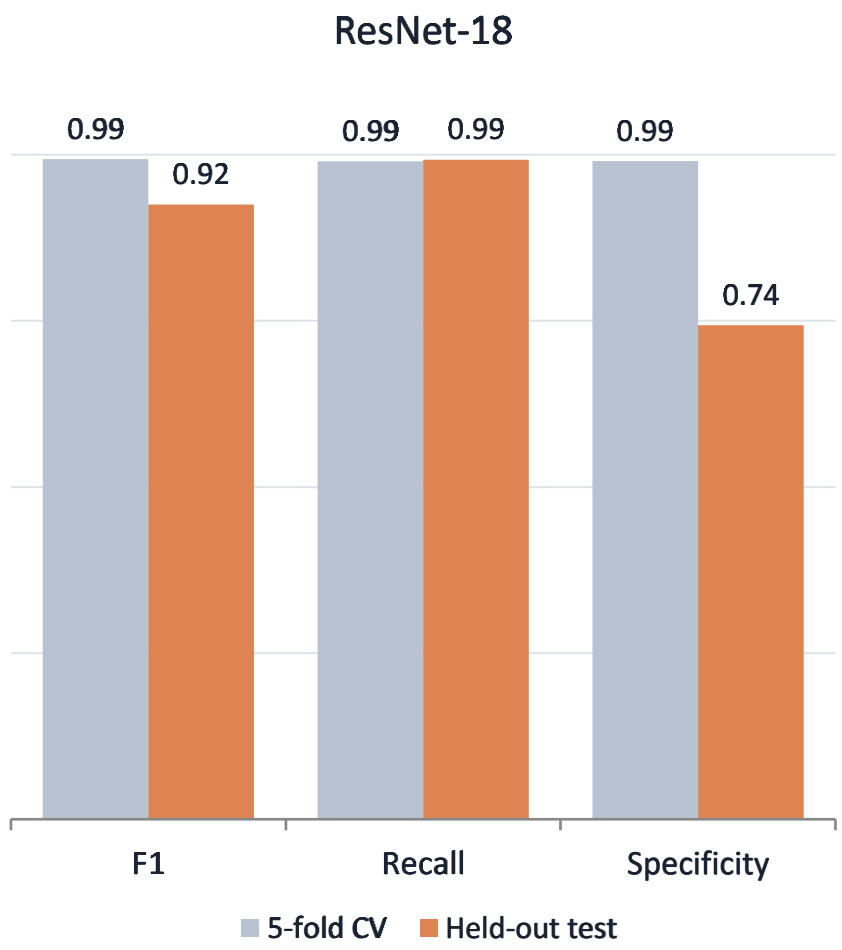
Held-out test: 624 images · 428 other children
Never used for training, tuning or early stopping



Why validate?
Training loss → 0 while validation loss levels off higher: that gap is overfitting. Early stopping keeps the best epoch per fold.

Near-perfect in validation, not on new patients





Specificity

healthy children correctly cleared

0.99 → 0.74

ResNet-18

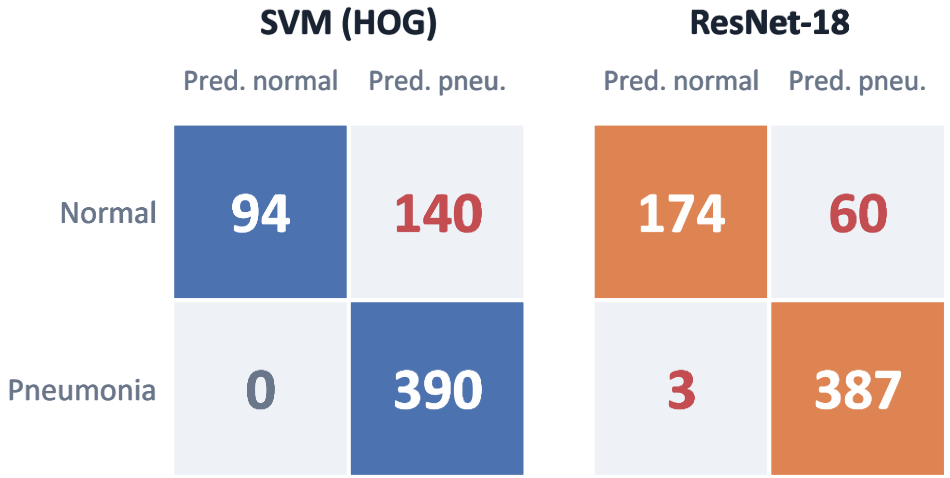
0.95 → 0.40

SVM (HOG)

CNN AUC 0.97: ranking survives; the 0.5 threshold doesn't. Most likely a distribution shift.

The CNN is significantly better overall

Held-out test (624 images)	SVM (HOG)	ResNet-18
Accuracy	0.78	**0.90**
Precision	0.74	**0.87**
Recall (sensitivity)	**1.000**	0.992
Specificity	0.40	**0.74**
F1	0.85	**0.92**
ROC-AUC	0.94	**0.97**
Missed pneumonia (of 390)	**0**	3
False alarms (of 234 healthy)	140	**60**



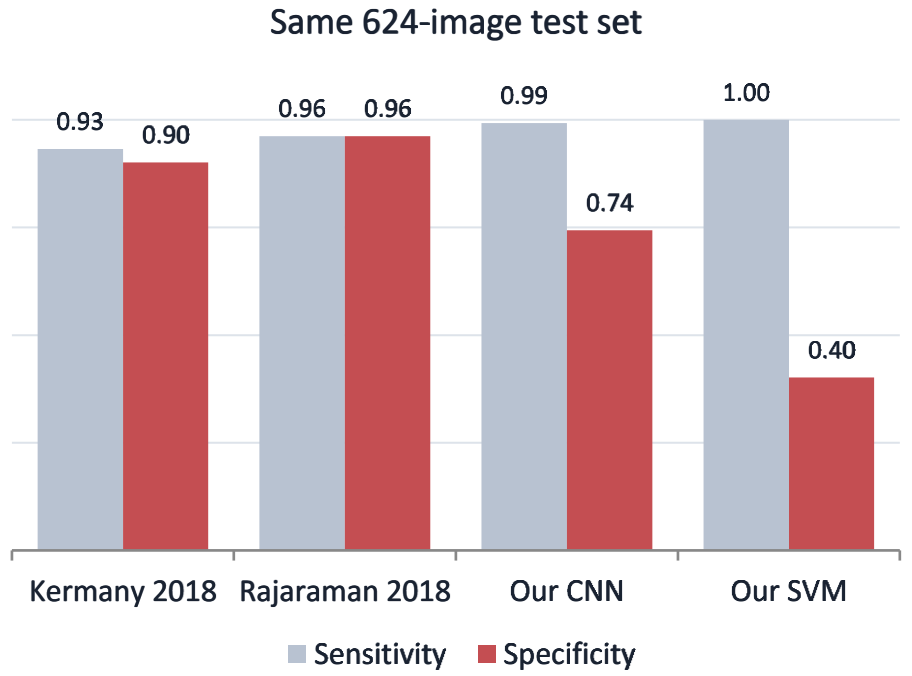
McNemar's exact test (paired)

82 vs **5**

test images only the CNN / only the SVM classified correctly · $p \approx 10^{-19}$

Trade-offs beyond accuracy, and the bigger picture

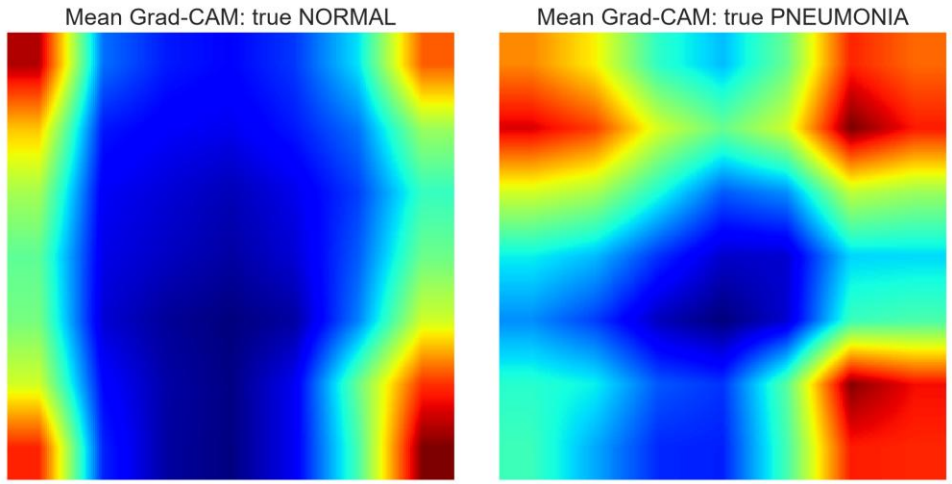
SVM vs CNN	SVM (HOG)	ResNet-18
Test specificity	0.40	**0.74**
Specificity, brighter images (γ = 0.5)	0.25	**0.59**
Explaining it	LIME only (opaque after PCA + RBF)	**Grad-CAM + LIME**
Data and pre-training	**None: learns from 5k images**	Needs ImageNet weights
Final training	**14 s on CPU**	26 s on GPU
Serving	**3.1 ms CPU · 7.4 MB**	13 ms CPU · 44.8 MB



Not tried here: random forest or logistic regression on the same HOG features; DenseNet-121 (CheXNet-style); segmenting the lungs before classifying.

At Kermany's sensitivity, our CNN ≈ 0.91 specificity. Similar ranking (AUC), different operating point. Threshold picked on test data: illustration only.

Grad-CAM: the CNN often looks at corners and labels

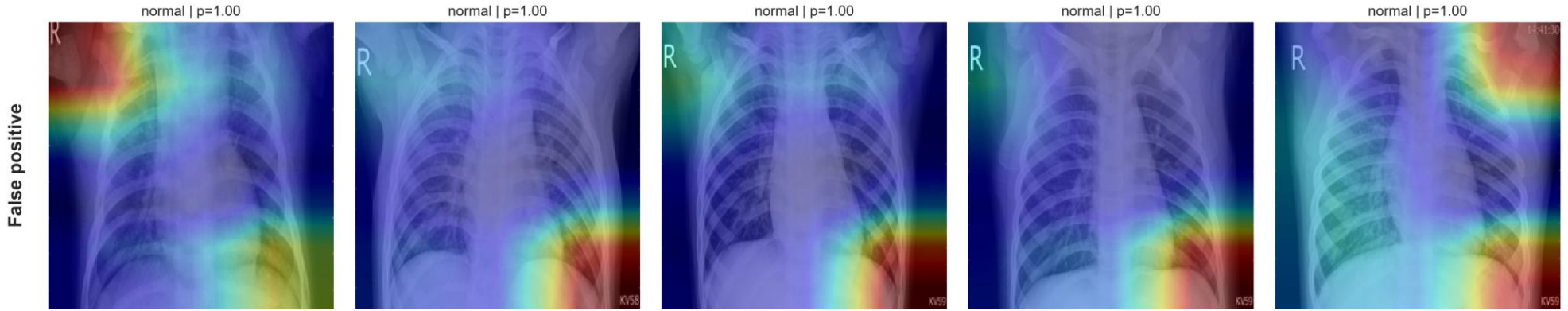


42%

of attention on flagged images falls in the outer border band (36% if spread evenly)

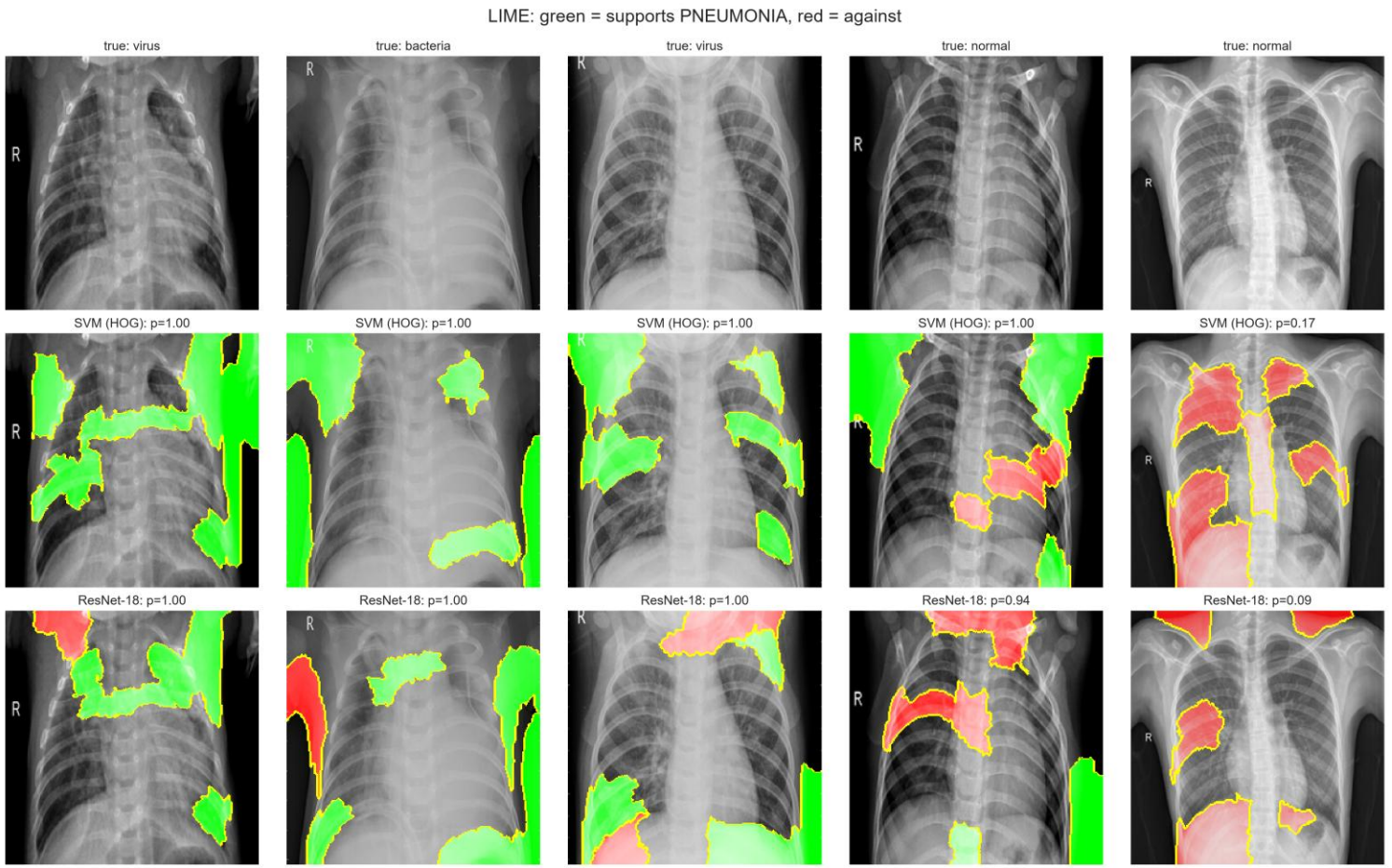
Shortcut learning: in all five of its most confident false alarms, the heat sits on burnt-in text: the 'KV58/KV59' exposure label (3), a timestamp (1) or the 'R' marker (1). A known failure mode (Zech et al. 2018; DeGrave et al. 2021).

The CNN's five most confident false alarms: healthy children, held-out test set



LIME: neither model reasons like a radiologist

Rows: original · SVM · ResNet-18 **Green** = pushes towards pneumonia **Red** = pushes away



Model-agnostic

Same perturbation method for both models: a fair comparison

Background matters

Influential patches often cover borders, abdomen and shoulders

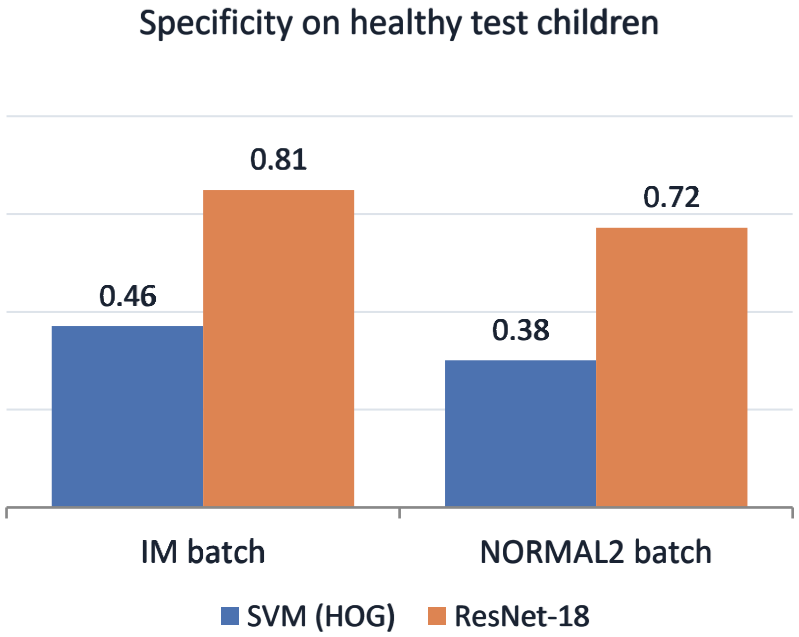
SVM sees edges anywhere

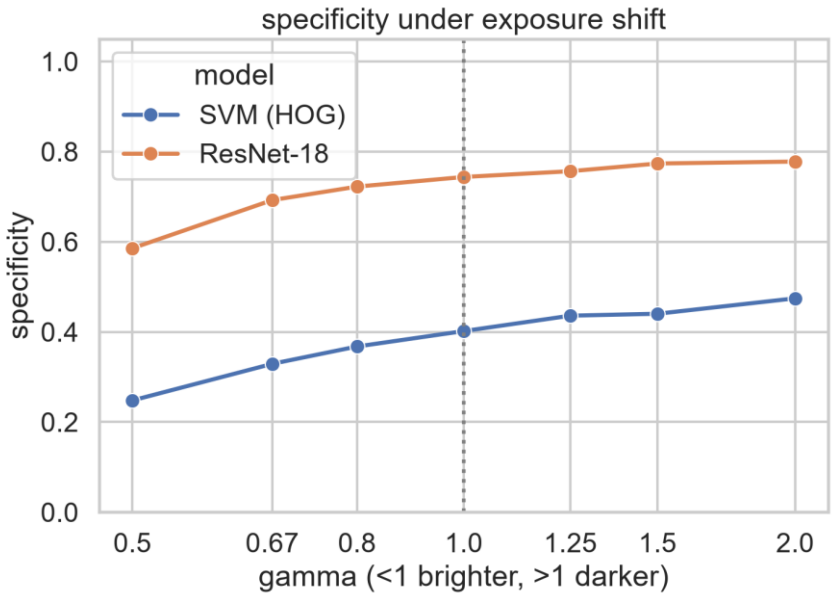
HOG has no notion of 'lung'

Unstable

Random perturbations; only five images, so read alongside Grad-CAM

Error rates depend partly on how the image was taken





No subtype gap

Bacterial and viral recall ≈ 1.0 for both models

Batch trend

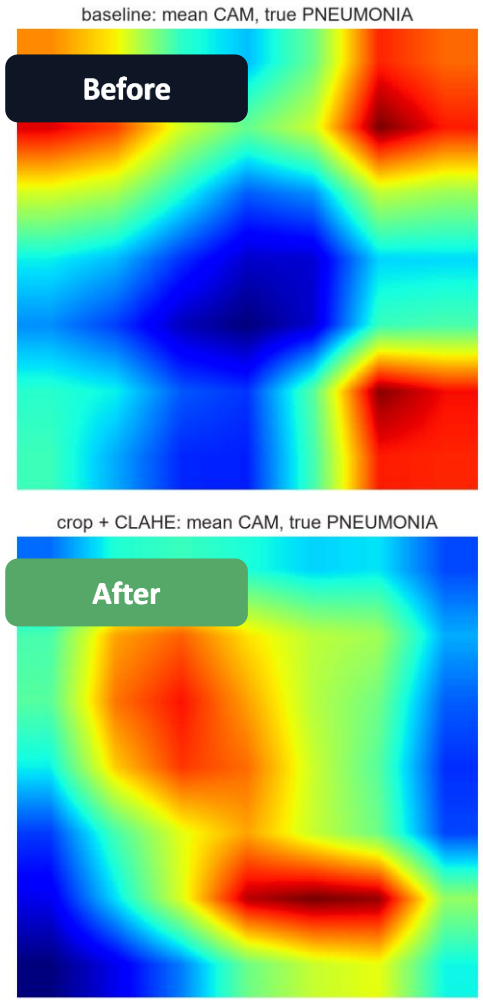
CNN clears 81% (IM) vs 72% (NORMAL2); intervals overlap

Brighter = more false alarms

Specificity 0.74 → 0.59 (CNN), 0.40 → 0.25 (SVM) at γ = 0.5

Real-world meaning: a hospital whose images differ in equipment, exposure or processing could see far more healthy children flagged, so the harm would fall unevenly across sites. **(71% of healthy test images are NORMAL2, against 56% in training.)**

Fixing where it looked made false alarms worse





Attention moved to the lungs

The mean heatmap shifts from the corners to the lung fields

More false alarms

CNN: 60 → 91 (McNemar $p \approx 10^{-6}$); no more robust to brightness

Here, the shortcut helps

Test images share this hospital's labels; only a new site can show which model generalises

A plausible explanation ≠ a better model

What the errors mean for a child

Data privacy

- Routine-care images; ethics approval and de-identification as reported by the authors
- No individual consent for ML reuse
- Burnt-in timestamps and codes: scrub pixels and DICOM headers

Errors are not equal

Missed pneumonia

Delayed antibiotics: potentially fatal

False alarm

Unneeded antibiotics, resistance, anxiety, workload

49% of CNN flags are real pneumonia at 20% prevalence (SVM: 29%)

Bias and harm

- One hospital, ages 1–5; no sex or ethnicity data, so fairness across them is untestable
- Batch and exposure change error rates
- Automation bias: a heatmap can falsely reassure

Recommendation: the CNN, as decision support only

RECOMMENDED

ResNet-18 CNN (baseline)

- Specificity 0.74 vs 0.40 for the SVM (McNemar $p \approx 10^{-19}$)
- Recall 0.992: 387 of 390 pneumonia cases caught
- Stays ahead on brighter images ($\gamma = 0.5$: 0.59 vs 0.25 for the SVM)
- About 13 ms per image on CPU: no GPU needed to serve

Known risk: it partly relies on burnt-in text. Mask text at de-identification, retrain on masked images, validate at each new site.

Only with…

Local calibration
Threshold re-set on the deploying hospital's data

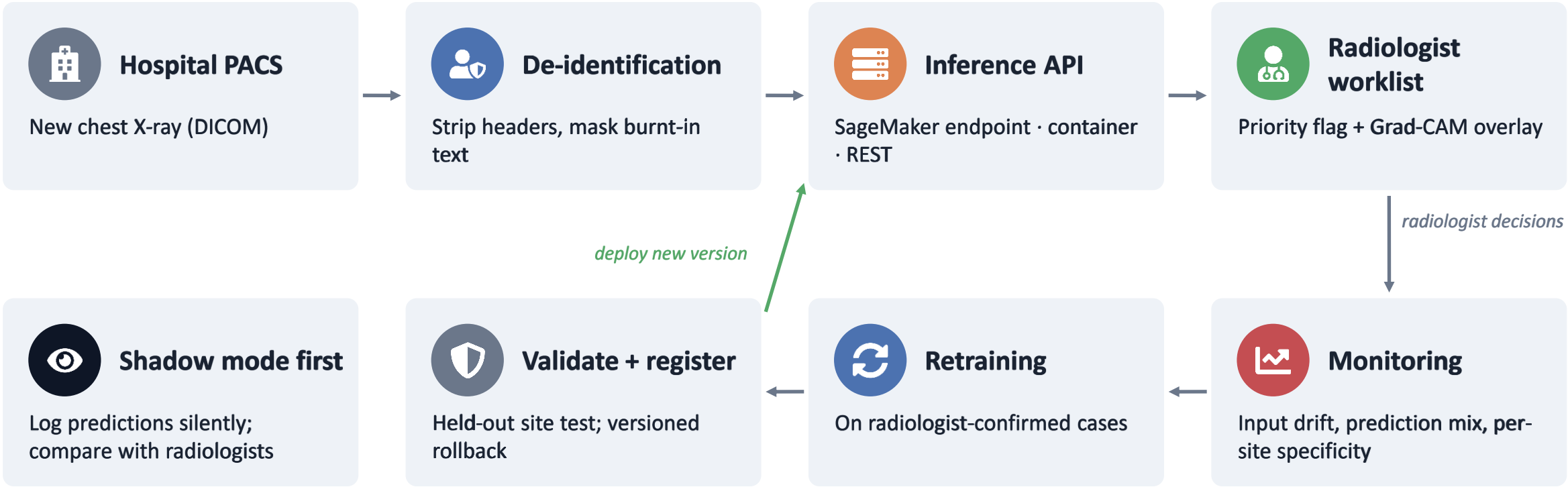
External validation
Tested at each new site before go-live

Shadow mode first
Predictions logged, not shown, until proven

Human in the loop
The radiologist makes every final decision

Regulatory approval
Software as a medical device (MHRA / UKCA)

Deploying it safely: human in the loop

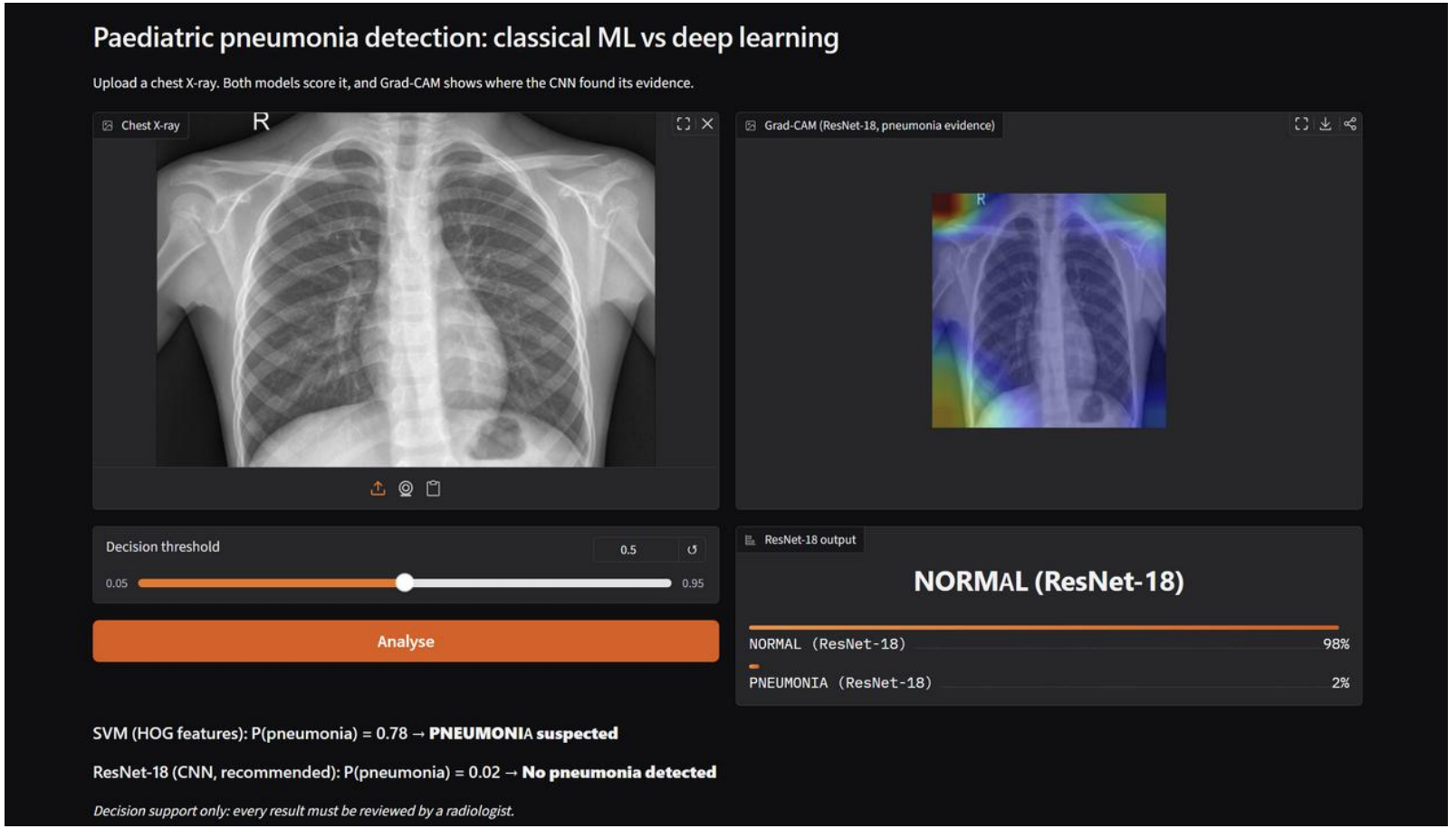


Scalability: stateless container, autoscaling CPU instances (≈ 13 ms/image)

Security: encrypted, in the hospital's UK/EU cloud region, full audit log

Platform: AWS SageMaker (or Azure ML, Vertex AI)

Demo: both models on an unseen X-ray



1. **Healthy child, unseen**

   SVM flags it; the CNN correctly clears it

2. **Viral pneumonia case**

   Both flag it; the heatmap shows the CNN's evidence

3. **Raise the threshold**

   The SVM's false alarm clears, at a cost in missed cases

```
Run: python demo/app.py
```

CONCLUSION

Key takeaways

1 The CNN wins here

Significantly better than the SVM (McNemar $p \approx 10^{-19}$); inference cost still negligible

2 Validation must mirror deployment

Patient-grouped CV overstated specificity by 25 (CNN) and 55 (SVM) points

3 XAI finds shortcuts, not fixes

Grad-CAM revealed label shortcuts; making the heatmaps look right made false alarms worse

4 Deploy only as decision support

Local calibration, external validation, shadow mode, a radiologist in the loop

If I did it again: a second hospital's data from the start · threshold set on validation data · nested CV and several random seeds · lung segmentation instead of a fixed crop

References

Cortes, C. and Vapnik, V. (1995) Support-vector networks. Machine Learning, 20, 273–297.

Dalal, N. and Triggs, B. (2005) Histograms of oriented gradients for human detection. CVPR, 1, 886–893.

DeGrave, A. J., Janizek, J. D. and Lee, S.-I. (2021) AI for radiographic COVID-19 detection selects shortcuts over signal. Nature Machine Intelligence, 3, 610–619.

Geirhos, R. et al. (2020) Shortcut learning in deep neural networks. Nature Machine Intelligence, 2, 665–673.

He, K. et al. (2016) Deep residual learning for image recognition. CVPR, 770–778.

Kermany, D. S. et al. (2018) Identifying medical diagnoses and treatable diseases by image-based deep learning. Cell, 172(5), 1122–1131.e9. Data: Mendeley Data, doi:10.17632/rscbjbr9sj.2

Loshchilov, I. and Hutter, F. (2019) Decoupled weight decay regularization. ICLR.

McNemar, Q. (1947) Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika, 12, 153–157.

Paszke, A. et al. (2019) PyTorch: an imperative style, high-performance deep learning library. NeurIPS.

Pedregosa, F. et al. (2011) Scikit-learn: machine learning in Python. JMLR, 12, 2825–2830.

Rajaraman, S. et al. (2018) Visualization and interpretation of convolutional neural network predictions in detecting pneumonia in pediatric chest radiographs. Applied Sciences, 8(10), 1715.

Ribeiro, M. T., Singh, S. and Guestrin, C. (2016) "Why should I trust you?" Explaining the predictions of any classifier. KDD, 1135–1144.

Selvaraju, R. R. et al. (2017) Grad-CAM: visual explanations from deep networks via gradient-based localization. ICCV, 618–626.

World Health Organization (2022) Pneumonia in children: fact sheet, 11 November 2022. who.int/news-room/fact-sheets/detail/pneumonia

Zech, J. R. et al. (2018) Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs. PLOS Medicine, 15(11), e1002683.

Zuiderveld, K. (1994) Contrast limited adaptive histogram equalization. In Heckbert, P. (ed.) Graphics Gems IV, 474–485.